

ECEN 4517 - PV Power Lab
Experiment 3 Part 2 Report
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1 Battery current and Voltage sensing

1.1 Schematic of sensing circuit

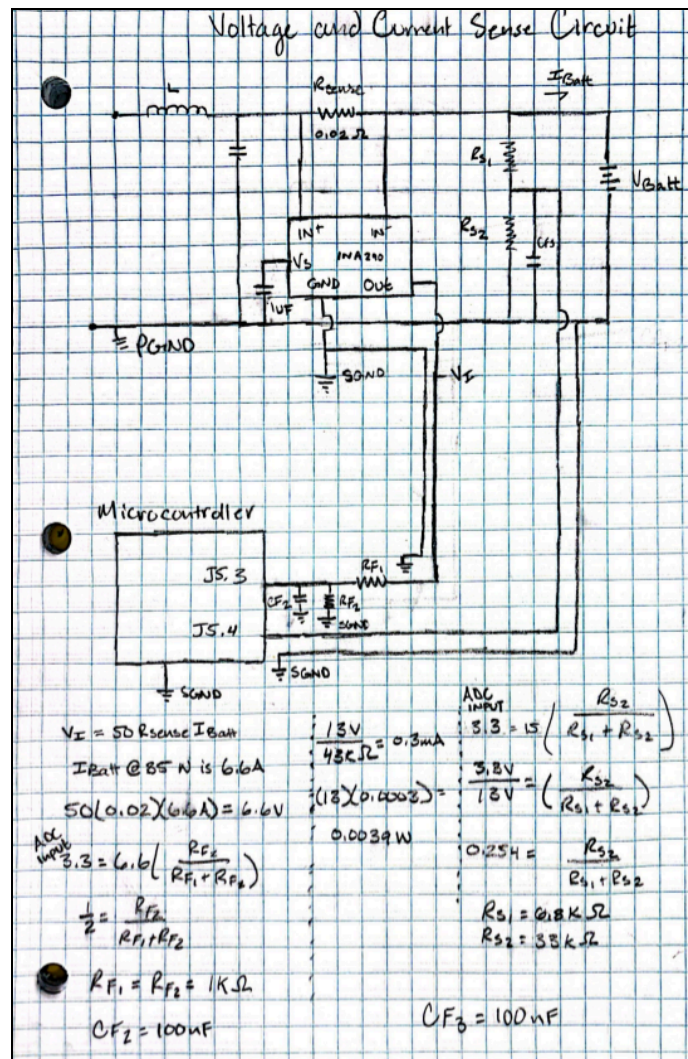


Figure 1.1 - 1: Sensing Circuit Schematic

Figure 1.1 - 1 shows expressions and calculations for the component values and voltages within the sensing circuit. R_{s1} and R_{s2} were chosen to be in the kiloOhm range to ensure very little current went through these resistors, but the voltages could still be used. A voltage divider equation was used to step down the maximum voltage of 15V to 3.3V. These values were taken from the Experiment 3 data in which we measured around 14V to be the maximum output of the buck converter, but wanted to ensure no components could possibly get damaged. R_{f1} and R_{f2} were chosen to be smaller than the other resistor pair since these were in series with a current sensor. The ratio was determined to be $\frac{1}{2}$, thus allowing R_{f1} to be identical to R_{f2} . The capacitor was chosen to be 100nF as this was a good value that was present in our lab kit. It should be mentioned that all component values were chosen to match what was available in the lab kit.

2 Testing of battery current and voltage sensing

2.1 Sensing circuitry

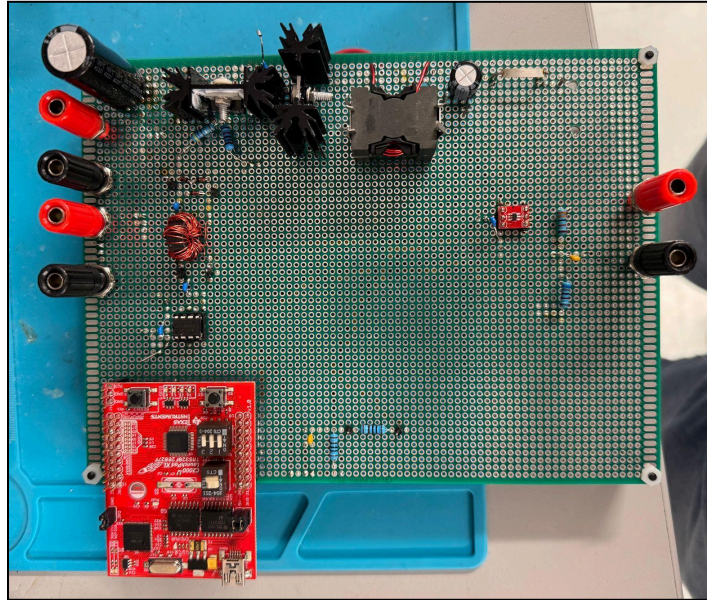


Figure 2.1 - Top View

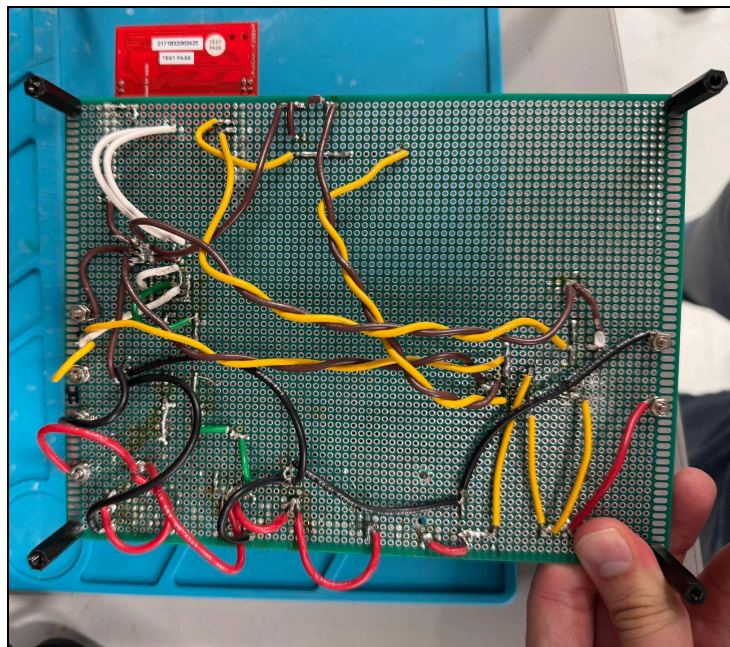


Figure 2.2 - Bottom View

2.2 Testing of Sensing Circuitry

For testing of the sensing circuitry, a while loop was used to take measurements of pins J5.3 and J5.4 (sensed_voltage and sensed current, respectively). In this loop, the sensed values were converted to real voltages and currents based on the resistor ratio used to create them. After some initial tuning, an averaging algorithm was written to ensure stable outputs for maximum power point tracking. The below graphs were created by slowly increasing the input voltage of the buck converter and measuring the output voltage and current on a set of multimeters. The multimeter data was plotted against the computer - sensed data to determine error in measurements.

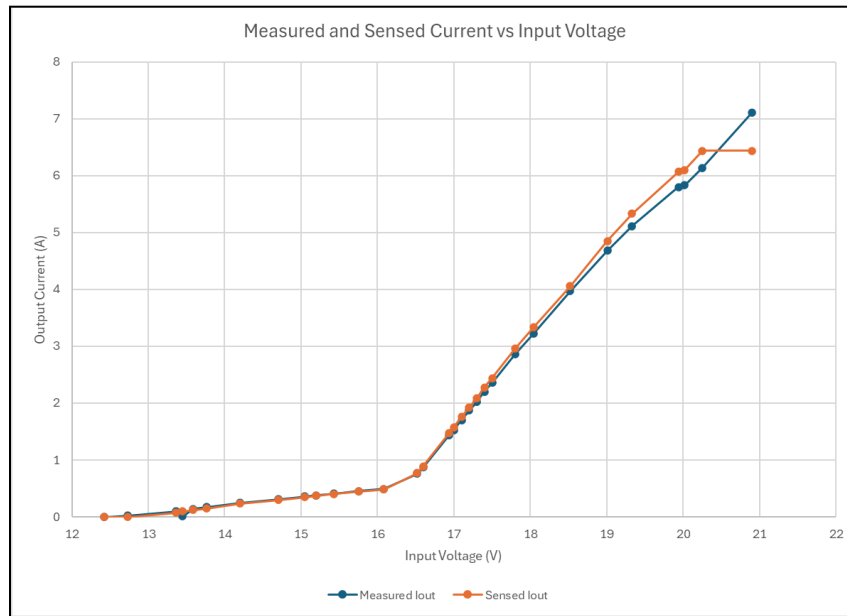


Figure 2.3 - 1: Measured and Sensed Output Current vs Input Voltage

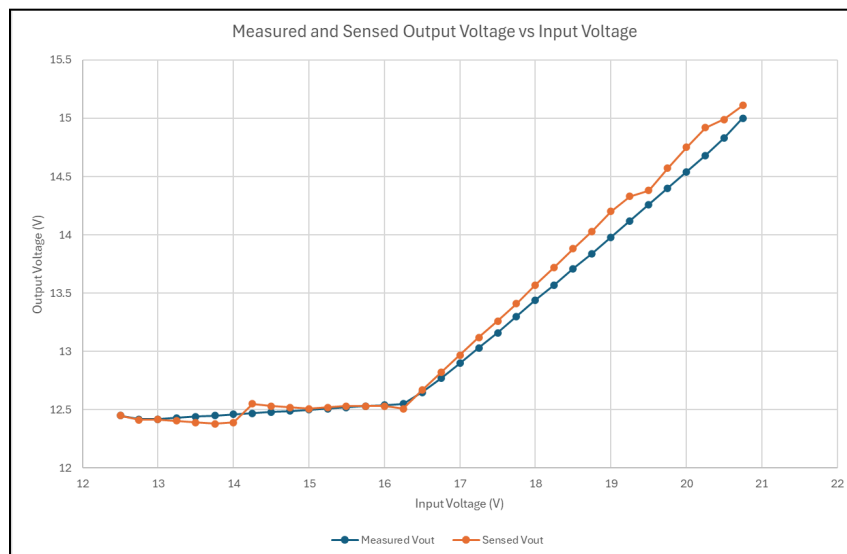


Figure 2.3 - 2: Measured and Sensed Output Voltage vs Input Voltage

The goal of Figures 2.3 - 1 and 2 was to determine how accurate the sensing circuitry was. As seen in the above plots, the sensing circuitry was able to almost perfectly match the multimeter measured values of the output current and output voltage. The voltage sensing circuitry had an average error of 0.8% in determining the actual voltage and the current sensing circuitry had an average error of 0.68% in determining the actual current.

3: MPPT and charge-control development and bench testing

3.1: MPPT Algorithm

For the MPPT algorithm, a duty cycle sweep was used to find the maximum power point. The algorithm can be broken down into 3 major sections with all but one taking place within the ADC interrupt. The first section is variable initialization, followed by initial measurements, and finally, maximum power point setting. Figure 3.1 - 1 shows the code used in this section.

```

17  interrupt void adc_isr(void);           // ADC interrupt service routine invoked after
18
19  float voltage = 0, power = 0, current = 0, duty = 0;           //variables for MPPT
20  float MPPT_Power = 0, MPPT_Duty = 0, MPPT_Current = 0, MPPT_Voltage = 0;           //variables for MPPT
21  float Rsense = 0.02;
22  float adjustedVoltage = 0;
23
24  unsigned int sensed_voltage,sensed_current;           // 12-bit ADC output (0-to-4095 corresponds to
25  unsigned int pulse_width;           // PWM pulse width expressed as a count of (1/
26  unsigned int period;           // PWM period expressed as a count of (1/60 MHz
27
28  int time;
29
30  const float TURNOFF_VOLTAGE = 14.15;
31  const float TURNON_VOLTAGE = 12.2;
32  const float MPPTCurrent = 5.178;
33  float voltSum = 0;
34  float currentSum = 0;
35  int countSum = 0;
36  float averageVoltage = 0;
37  float averageCurrent = 0;
38  const long int SECOND = 4325883;

```

Figure 3.1 - 1 MPPT Variable Declarations

Above shows the different variables that are used within the MPPT algorithm. The “MPPT_x” variables are the maximum power point variables that hold the current, voltage, duty cycle and power measurements at peak power. “current”, “voltage”, “power”, and “duty”, show the real time measurements of the converter. These will be checked every duty cycle against the MPPT variables. The “sensed_voltage” and “sensed_current” variables are the 12 bit ADC values coming directly from the sensors. These values are converted into voltages and currents that are used later in the algorithm.


```

143 interrupt void adc_isr(void)
144 {
145     if (averageVoltage >= 12.6) // dont start the charging until it drops below 12.6
146     {
147         EALLOW;
148         EPwm2Regs.CMPA.half.CMPA = (period)>>1; // Stars the sampling of the current and voltage s
149         EDIS; // This could be changed to whatever the student r
150
151         AdcRegs.ADCINTFLGCLR.bit.ADCINT1 = 1; // Clear ADCINT1 flag reinitialize for next SOC
152         PieCtrlRegs.PIEACK.all = PIEACK_GROUP1; // Acknowledge interrupt to PIE
153         return;
154     }
155
156     // this is essentially the else condition, run the mppt if we're below 12.6V
157     long int i = 0;
158     MPPT_Current = 0;
159     MPPT_Duty = 0;
160     MPPT_Power = 0;
161     MPPT_Voltage = 0;
162
163     for (duty = 0.55; duty < 0.95; duty += 0.025)
164     {
165         pulse_width = (duty * period);
166         update_duty(pulse_width);
167
168         for (i = 0; i < (SECOND * 0.5); i++)
169         {
170
171
172
173         while (countSum <= 300) // loop to average the measurements
174         {
175             sensed_voltage = AdcResult.ADCRESULT1; // Voltage sensed on pin J5.3 of t
176             sensed_current = AdcResult.ADCRESULT0; // Voltage sensed on pin J5.4 of t
177
178             current = ((float)sensed_current/4095);
179             current = ((6.79 * current)/ (50 * Rsense));
180             voltage = ((float)sensed_voltage / 4095);
181             voltage = (voltage * 19.51);
182             countSum++;
183             voltSum += voltage;
184             currentSum += current;
185
186             if (countSum == 300)
187             {
188                 averageVoltage = (voltSum / (float)countSum);
189                 averageCurrent = (currentSum / (float)countSum);
190                 countSum = 0;
191                 voltSum = 0;
192                 currentSum = 0;
193                 break;
194             }
195         }
196     }

```

Figure 3.1 - 2: Initial Measurements

Figure 3.1 - 2 shows the first operations within the MPPT algorithm. This section of code is the initial measurements taken from the Analog to digital converters. Lines 147 to 153 execute the ADC Interrupt service routine, if we are above the turn on voltage of 12.6V. This would mean that the battery is at a good state of charge and the buck converter can remain turned off. If this returns false, then the code progresses to the next section which is taking measurements. Lines 163 - 193 show a large “For” loop in which the duty cycle is swept from 55% to 95% in 2.5% increments. Within the larger “For” loop is a secondary smaller “for” loop which is used to wait for the converter to achieve steady state before taking any measurements. The macro, “SECOND”, is 4,325,883, which is the number of cycles needed for roughly 1 second of time to pass. This value was experimentally determined with initial basis off the internal clock running at 60MHz. After the system has achieved steady state, the ADC performs 300 measurements of the current and voltage (lines 171 - 193) and averages these values in order to obtain the “averageCurrent” and “averageVoltage” values.

Below, Figure 3.1 - 3, shows the MPPT calculations that occur after the initial measurements. The variables “averageCurrent” and “averageVoltage” are first adjusted down by 1% if we are in a high duty cycle. This decision was influenced by the initial data gathered in Section 2.2. That data showed that the sensed voltage and current were above what the actual values should have been at high input voltages. Therefore, a small 1% adjustment downward should make these values more accurate. After the adjustments, a power calculation is made and all values are compared against the MPPT variables (lines 207 - 212). After all the duty cycles have been swept, the maximum power point duty cycle is then used for the next 5 minutes while the rest of the system is told to sleep. After each minute, another measurement is taken to ensure that the battery does not get overcharged. If the battery reaches 12.8V or the five minutes are over, the system restarts the duty cycle sweeping process.


```

200     if (averageVoltage > 14)
201     {
202         averageCurrent = (averageCurrent * 0.99);
203     }
204
205     power = averageCurrent * averageVoltage;
206
207     if (power >= MPPT_Power)
208     {
209         MPPT_Power = power;
210         MPPT_Duty = duty;
211         MPPT_Current = averageCurrent;
212         MPPT_Voltage = averageVoltage;
213     }
214 }
215
216     update_duty(period * MPPT_Duty);
217     int j = 0;
218     for (j = 0; j < 5; j++) // run for 5 minutes before next duty sweep. this is the
219     {
220         for (i = 0; i < (SECOND * 60); i++) // this is a 60 second loop - ish
221         {
222         }
223         // after each minute, check if we need to shut off.
224         while (countSum <= 100) // loop for finding voltage
225         {
226             sensed_voltage = AdcResult.ADCRESULT1; // Voltage sensed on pin J5.3
227             voltage = ((float)sensed_voltage / 4095);
228             voltage = (voltage * 19.51);
229             countSum++;
230             voltSum += voltage;
231
232             if (countSum == 100)
233             {
234                 averageVoltage = ((voltSum / (float)countSum) * 0.99);
235                 countSum = 0;
236                 voltSum = 0;
237                 break;
238             }
239         }
240
241         if (averageVoltage > TURNOFF_VOLTAGE) // determine shutoff
242         {
243             update_duty(0); // shut off system.
244             break;
245         }
246     }
247
248     EALLOW;
249     EPwm2Regs.CMPA.half.CMPA = (period)>>1; // Starts the sampling of the current
250     EDIS; // This could be changed to whatever
251
252     AdcRegs.ADCINTFLGCLR.bit.ADCINT1 = 1; // Clear ADCINT1 flag reinitialize
253     PieCtrlRegs.PIEACK.all = PIEACK_GROUP1; // Acknowledge interrupt to PIE
254     return;
255 }

```

Figure 3.1 - 3: MPP Setting Code

3.2 In Lab Testing of MPPT Algorithm

Before the buck converter and the MPPT system were taken outside, initial testing was conducted within the lab to ensure proper functionality. The system was connected to two bench top power supplies, a rheostat, and the battery for this experiment. The rheostat was set to 3.4 Ohms to simulate the input resistance of the PV Panel, R_{in} . This resistance was found using the below equation.

$$\frac{Max\ PV\ Power}{(Max\ PV\ Current)^2} = Input\ Resistance \quad \frac{85W}{(4.94\ A)^2} = 3.48\ \Omega$$

Since the system was designed to have as few losses as possible, the same equation was used to calculate the output resistance of the battery based on previously acquired experimental data. With an output power of 72.76 W at 5.1 A, the output resistance was measured to be 2.8 Ω . The closer these resistance values are the fewer reflections will occur and fewer losses in the converter. With the resistances measured and set, a voltage divider had essentially been created. The input voltage set on the bench top power supply was then set to 38V in order to achieve the optimal 17.2V at the input of the converter. This was calculated using the following equation.

$$38V \frac{2.8\ \Omega}{2.8\Omega + 3.4\Omega} = 17.16\ V$$

The MPPT code and testing was broken into parts to ensure no components were damaged in the process. The first step was to test if the sensing circuitry was still reading the correct values at a single duty cycle. After this was verified, the duty cycle was then swept within a for loop, and the sensing circuitry was double checked again. Then power calculations were introduced to determine maximum power point tracking and to test the full system. The MPPT variables were watched to determine if they were changing as the duty cycle increased and to see when they stopped changing. The below table shows the data collected both on a series of multimeters and by the sensing circuitry. The sensing circuitry, in blue, was able to find the maximum power point repeatedly. While the true values of the power were off by 5W, this showed great promise that the MPPT algorithm was able to detect increases and decreases in the overall output power of the system. The converter consistently found that 74.7 W was the maximum power and operated at an efficiency of 93%.

	Cycle Number	Input Voltage	Input Current	Output Voltage	Output Current	Input Power	Output Power	Efficiency
MultiMeter	1	17.0	4.7	13.7	4.4	79.9	60.3	75.4
	2	17.1	4.7	14.0	5.3	80.4	74.2	92.3
	3	17.2	4.7	14.1	5.3	80.8	74.7	92.4
	4	16.7	4.8	14.1	5.3	80.2	74.7	93.2
	5	17.5	4.5	14.0	5.3	78.8	74.2	94.2
	6	17.1	4.7	14.0	5.3	80.4	74.2	92.3
	7	17.1	4.7	14.1	5.3	80.4	74.7	93.0
	8	17.1	4.7	14.1	5.3	80.4	74.7	93.0
Computer	1	17.1	4.7	14.2	6.1	80.6	86.6	107.5
	2	17.1	4.8	14.1	5.6	78.7	79.5	101.1
	3	17.2	4.6	14.2	5.6	79.1	79.6	100.6
	4	17.1	4.7	14.2	5.6	80.4	79.5	98.9
	5	17.1	4.7	14.3	5.5	80.4	78.7	97.9
	6	17.2	4.7	14.2	5.5	80.8	78.7	97.4
	7	17.4	4.6	14.1	6.1	80.0	78.7	98.3
	8	17.0	4.7	13.9	5.6	79.9	78.7	98.5

Table 3.2 - 1: MPPT Data During Benchtop Testing

Below is Figure 3.2 - 1, which shows the Maximum Power Point Tracking Algorithm working over time. The y-axis represents the logged maximum power found, and the x-axis represents time. Based on the graph, the MPPT algorithm found a local maximum of roughly 72W around an early duty cycle and then proceeded to finding higher and higher powers with increasing duty cycles. It eventually found the maximum power point at a duty cycle of 95%. This is not represented in the graph, but a note was made during the time of testing.

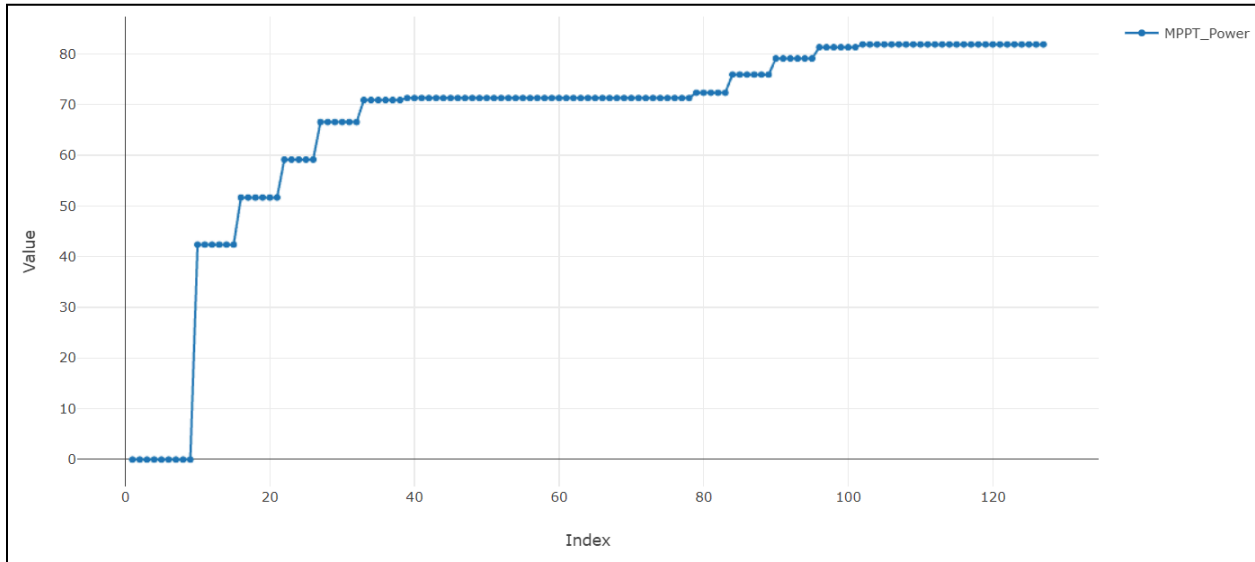


Figure 3.2 - 1: Maximum Power Point Tracking

Below in Figure 3.2-2 is our charge-control feature. This feature shuts off the converter if the battery voltage exceeds 12.8V and turns the system back on if the battery voltage drops below 12.6V. While the figure below shows voltages above those values, this is due to the offset in the voltage sensing circuit.

Notes and data were taken during the initial testing of the sensing circuit, and the turnoff and turnon values were adjusted accordingly. At the 450 second mark, the program turns off the system. The turnoff check happens at the end of the program and after a few delays which is why this voltage is set for over 2 minutes.

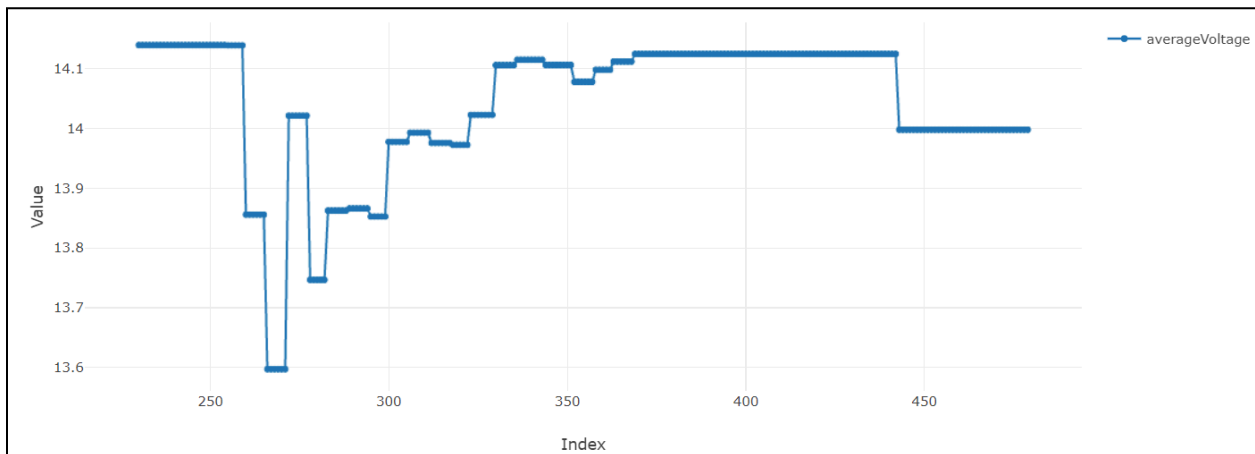


Figure 3.2 - 2: MPPT Turn Off Feature

Below is Figure 3.2 - 3 which shows the filtered Vpwm out of J6.1 of the Launch Pad MicroController in blue, and Vin, the voltage waveform of the benchtop power supply in yellow. The filter chosen was a low pass filter comprised of a 150 Ohm resistor and a 100nF capacitor, giving a corner frequency of 10kHz. At 100Khz, there should be 20dB of attenuation since this is a first order filter. Based on the below screen capture, there is quite a bit of noise in both signals, specifically, when the switching of the PWM occurs. This is most likely due to the long wires on the board, which is susceptible to switching noise when there is high changing in current, per change in time. This same filter is used in later sections, specifically, during shaded and unshaded testing of the system with the PV panel.

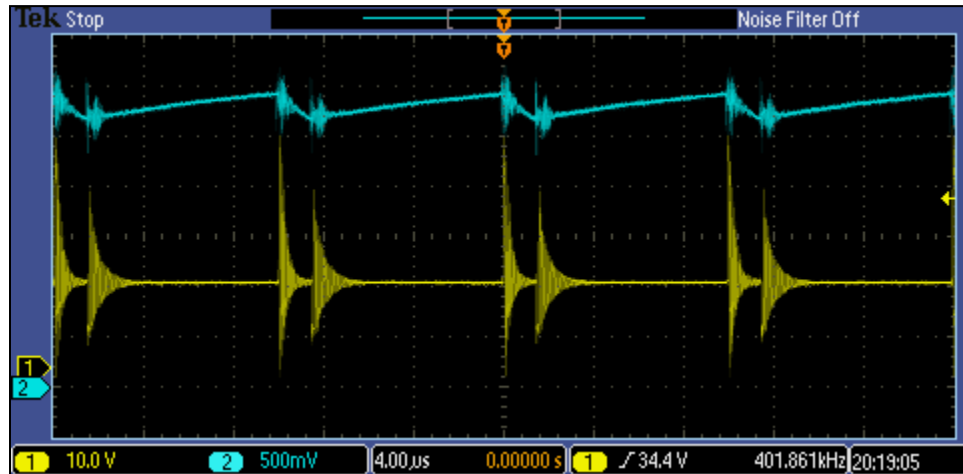


Figure 3.2 - 3: Filtered Vpwm and Vin

4 Complete System Testing

4.1 Unshaded Solar Testing

After completing benchtop testing of the system, the buck converter was taken outside to be tested with the PV Panel. The initial test was ran at 1:39pm on Saturday March 8th, 2025. The solar irradiance was measured at 1160 W/m². The below table shows the measured maximum power point found by the system.

Unshaded							
Vin	Iin	Pin	Vout	Iout	Pout	Efficiency	Duty Cycle
14.805	4.840	71.656	13.705	5.100	69.896	97.5	95
14.540	4.590	66.739	13.100	4.950	64.845	97.2	95
14.801	4.807	71.148	13.699	5.047	69.139	97.2	95
14.212	4.747	67.464	13.526	4.879	65.993	97.8	95
14.693	4.760	69.939	13.604	5.006	68.102	97.4	95
14.691	4.748	69.753	13.597	5.001	67.999	97.5	96
14.729	4.771	70.272	13.629	5.012	68.309	97.2	96
14.575	4.665	67.992	13.145	5.045	66.317	97.5	92

Table 4.1 - 1: Unshaded Maximum Power Point Data

Some notable points about this test was the state of the battery when taken out. The previous group to have used the PV cart did not charge the battery properly and was seen to have 12.04V at the start of testing. This corresponds to a state of charge below 50%. With the low state of charge and the load that was attached to the battery during testing, the turn off charging feature was not able to be tested. Even so, the system seemed to operate very well, finding an average maximum power point of 67.57W with an average input power of 69.37W. This corresponds to an average efficiency of 97.4%. While these power numbers are lower than what was anticipated, it shows great promise, seeing that the efficiency of the converter was so high.

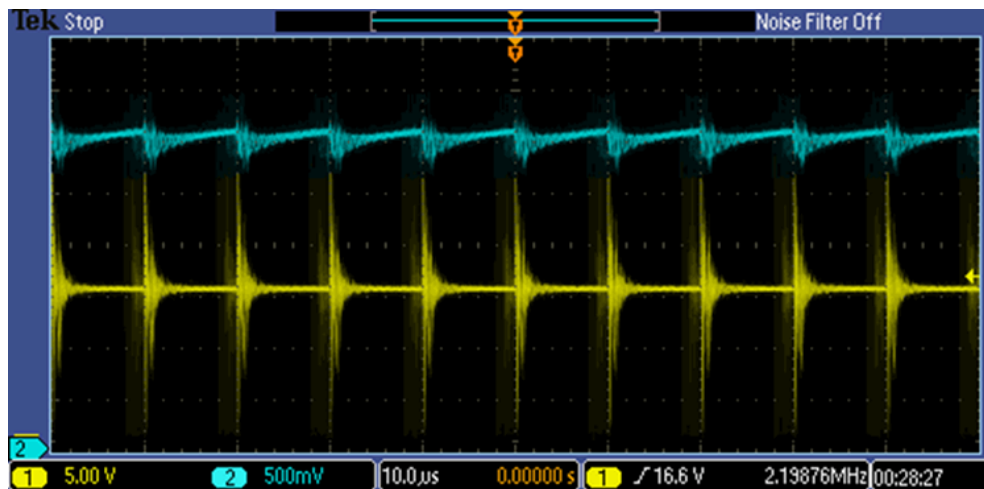


Figure 4.1 - 1: Low Pass Filtered Vpwm and Vpv

Figure 4.1 - 1 shows the voltage waveforms of the low pass filtered V_{pwm} , the output pwm in blue, and V_{pv} , the output voltage waveform of the PV panel in yellow. The filter that was selected, is the same that was used during the benchtop testing in section 3. This waveform was taken during the maximum power point cycle (5 minutes, see page 7 & 8).

4.2 Unshaded Data Comparison to Previous Testing

In Experiment 2, data was collected to characterize the PV Panel with a purely resistive load. This data featured measurements of power, current, and voltage for both a shaded and unshaded case.

Unshaded				Vin	I in	Pin
Voltage (cell) (V)	Voltage (Panel) (V)	Current (A)	Power (W)			
0.542	19.5	1.5	29.25	14.805	4.840	71.656
0.539	19.4	1.6	31.04	14.540	4.590	66.739
0.536	19.3	1.8	34.74	14.801	4.807	71.148
0.525	18.9	2.3	43.47	14.212	4.747	67.464
0.511	18.4	2.9	53.36	14.693	4.760	69.939
0.486	17.5	3.6	63	14.691	4.748	69.753
0.475	17.1	4	68.4	14.729	4.771	70.272
0.464	16.7	4.4	73.48	14.575	4.665	67.992
0.403	14.5	5.1	73.95			
0.306	11	5.3	58.3			
0.114	4.1	5.3	21.73			
0.197	7.1	5.3	37.63			
0.094	3.4	5.3	18.02			

Figure 4.2 - 1: Experiment 2 Data vs MPPT Data

Figure 4.2 - 1 shows the Experiment 2, purely resistive load data on the left (gray) and Experiment 3 MPPT data on the right (blue). During experiment 2, a maximum input power of 73.95W was found. In experiment 3, a maximum power of 71.656 W was found. The MPPT tracking was 3.2% away from the previously found maximum in experiment 2. While many things have changed since then, such as a buck converter, a different load, sensing circuitry, and different initial battery voltages. It is completely possible that the experiment 3 system is working perfectly and as expected.

Furthermore, in experiment 2, direct energy transfer data was collected to see how well the PV panel could charge the battery by itself.

Battery Voltage (V)	12.8
Battery Current (A)	5.0
Power (W)	64

Vout	I out	Pout
13.705	5.100	69.896
13.100	4.950	64.845
13.699	5.047	69.139
13.526	4.879	65.993
13.604	5.006	68.102
13.597	5.001	67.999
13.629	5.012	68.309
13.145	5.045	66.317

Figure 4.2 - 2: Direct Energy Transfer Comparison with MPPT Data

Figure 4.2 -2 shows the direct energy transfer data in white, and the MPPT data in red. It is quite apparent that the MPPT data, and the buck converter system, outperforms the direct energy transfer method. Using the average output power from the MPPT data, the system outputted 5.6% more power than the direct energy transfer method.

4.3 Shaded Solar Testing

After completing unshaded testing of the system, the PV Panel had 4 cells covered or blocked by a thick coat. The initial test was ran at 3:03pm on Saturday March 8th, 2025. The solar irradiance was measured at 1120 W/m². The below table shows the measured maximum power point found by the system.

Shaded							
Vin	I in	Pin	Vout	I out	Pout	Efficiency	Duty Cycle
12.461	0.254	3.16	12.127	0.262	3.17	100.4	96
12.425	0.217	2.69	12.099	0.223	2.70	100.2	96
12.344	0.124	1.53	12.088	0.133	1.61	105.1	95
12.057	0.644	7.76	12.200	0.676	8.25	106.2	95
13.970	0.614	8.58	12.180	0.700	8.53	99.4	87
12.850	0.681	8.75	12.190	0.716	8.73	99.7	95

Table 4.3 - 1: Shaded Solar MPPT Data

While the unshaded data shown in section 4.1 shows fairly consistent results, the shaded data taken seems to be more sporadic than anticipated. The average output power of the system was 5.5 W with an average efficiency of 101.8%. This shows that the system was not operating correctly and could be caused by the shading of the panel with a low state of charge in the battery. If the state of charge is low, then the battery would pull more current while remaining at a lower voltage. The shading of the PV Panel could have caused the MPPT algorithm to respond incorrectly. Overall, more testing is needed with the MPPT algorithm, specifically, in scenarios where the input voltage and current are low.

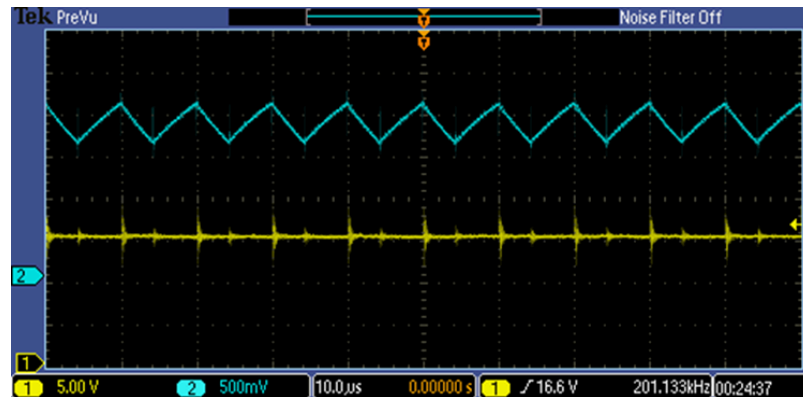


Figure 4.3 - 2: Vpwm and Vpv waveforms

Figure 4.3 - 2 shows the voltage waveforms of the PV panel, Vpv in yellow, and the filtered pwm, Vpwm in blue. The filter used was the same LPF from sections 4.1 and 3.2. It would seem, at lower PV voltages, there is dramatically less ringing and the pwm signal filtering produces a more triangular waveform, similar to a capacitor charging and discharging.

4.4 Shaded MPPT Data Comparison with Direct Energy Transfer

In Experiment 2, data was collected to characterize the PV Panel with a purely resistive load. This data featured measurements of power, current, and voltage for both a shaded and unshaded case.

4 Cells Shaded				Vin	I in	Pin
Voltage (cell) (V)	Voltage (panel) (V)	Current (A)	Power (W)			
0.261	9.4	0.8	7.52	12.461	0.254	3.16
0.258	9.3	0.9	8.37	12.425	0.217	2.69
0.256	9.2	1.14	10.49	12.344	0.124	1.53
0.253	9.1	1.3	11.83	12.057	0.644	7.76
0.251	9.05	1.47	13.30	13.970	0.614	8.58
0.250	9	1.5	13.50	12.850	0.681	8.75
0.244	8.8	1.8	15.84			
0.244	8.8	1.98	17.42			
0.232	8.34	2.84	23.69			
0.225	8.1	3.2	25.92			
0.186	6.7	4.8	32.16			
0.156	5.6	4.94	27.66			
0.058	2.1	5.1	10.71			
0.037	1.32	5.1	6.73			

Figure 4.4 - 1: Experiment 2 vs MPPT Solar Data

Figure 4.4 - 1 shows the Experiment 2, purely resistive load data, on the left in gray. The blue data on the right shows the MPPT data. According to the Experiment 2 data, the PV panel is capable of producing 32.16 W at its maximum power point. The MPPT Algorithm was only able to find a maximum input power of 8.75W which is dramatically lower than what was expected. This might be due to the state of charge of the battery, an error in the MPPT algorithm under shaded conditions, or incorrect measurement techniques during the time of the experiment. While the true source of the discrepancy is unknown, there is still validity to the data collected in the unshaded scenario. This system underwent scrupulous testing to ensure proper functionality of every component. At this time, more data and testing is needed to determine the route cause of the inconclusive results of this test.

For direct energy transfer, experiment 2 never performed a shaded direct energy transfer experiment. Therefore, there are no results to compare this experiment's data to. However, based on the Characterization data of the solar panel from experiment 2, a linear approximation can be made. The PV panel produced a maximum power point of 73.95W with a purely resistive load and 64W while charging the battery via the direct energy transfer method. Therefore, with a shaded maximum power point of 32.16W, it can be inferred that the PV panel would produce 27.83W via the direct energy transfer method. Given that the MPPT system only provided a max of 8.75W, the system was still well below what the direct energy transfer method would be able to provide. Once again, this ~3x discrepancy will require more testing and investigation to determine the route cause.